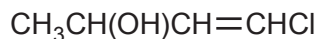


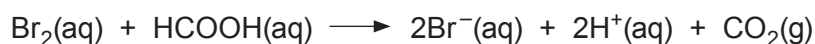
3. Draw a section of the addition polymer formed from the monomer below.



You should show **two** repeat units.

[1]

4. Bromine reacts with methanoic acid according to the equation below.



A student wanted to follow the rate of the reaction and mixed solutions of known concentration of bromine and methanoic acid.

Suggest **two** methods which the student could use to follow the rate of this reaction as it proceeds. [2]

Method 1

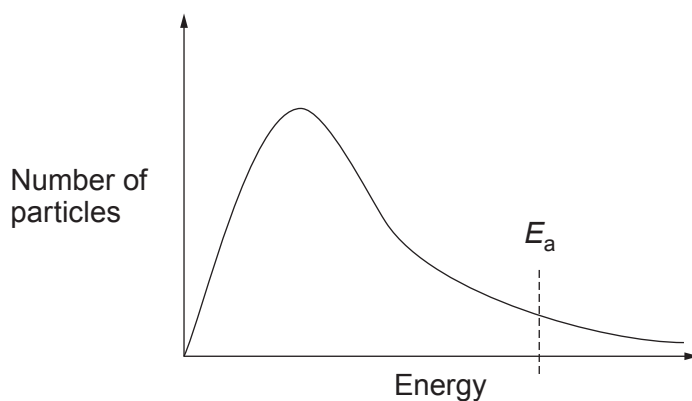
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Method 2

.....



3. The diagram shows the distribution of the energies of particles in a gas at a particular temperature.



On the diagram draw a curve to show the distribution of the energies of the same particles at a higher temperature.

Use the diagram to explain the effect of increasing temperature on the rate of a reaction. [2]

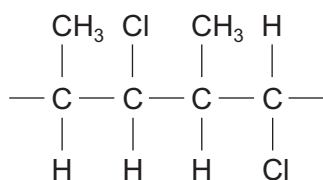
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03

4. A section of polymer is shown.



Draw the structural formula of the monomer used to make this polymer. [1]



3. A student suspects an unlabelled organic liquid is a carboxylic acid. Name the reagent(s) that must be added to the unknown organic liquid to test for the presence of a carboxylic acid. Give the expected observations for a positive result. [2]

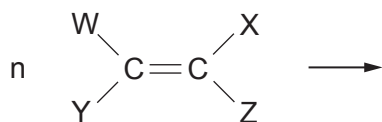
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4. Complete the equation below to show the product of addition polymerisation. [1]



5. State the meaning of the term 'standard enthalpy change of formation'. [2]

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